

Video Summary

This Telusko lesson introduces Python sets as collections of unique, unordered elements. It explains hashing, empty set creation with `set()`, fast membership checks, mathematical set operations, and a quick single-element tuple reminder.

Source video: <https://www.youtube.com/watch?v=-j9Q2PQvfM&list=PLsyebzWxl7omDoEYrrf3oXvXxa6MPgek&index=10>

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Defining Sets

A set is a collection of unique and unordered elements. Sets are commonly written with curly brackets when they already contain values.



Unique values

Duplicate items collapse into one stored value.



Unordered

Sets do not preserve the order you typed values in.



Mutable set

You can add and remove set items after creation.

Set definition examples

```
>>> nums = {25, 12, 36, 12, 25}
>>> nums
{25, 12, 36} # order may vary
>>> letters = set("python")
>>> "p" in letters
True
```

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Hashing Mechanism

Sets use hashing to place and find values quickly. This is why sets are fast for membership tests, but do not keep insertion order.

#

Hashing

A hash decides where a value is stored internally.

in

Fast lookup

Checking item in set is usually very quick.

?

No order

Printed output can appear in a different order.

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Creating Empty Sets

Use `set()` to create an empty set. Empty curly brackets create an empty dictionary, not an empty set.

Empty set examples

```
>>> empty_set = set()
>>> type(empty_set)
<class 'set'>
>>> empty_braces = {}
>>> type(empty_braces)
<class 'dict'>
```

Beginner rule

Use `set()` when you want an empty set. Use `{}` only when you want an empty dictionary.

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Common Set Operations

Python supports mathematical set operations with compact symbols. These operations return new sets.

Symbol	Name	Meaning	Example
&	Intersection	Only values common to both sets.	a & b
-	Difference	Values in the first set but not the second.	a - b
	Union	All unique values from both sets.	a b
^	Symmetric diff	Values that are not common to both sets.	a ^ b

Numeric set example

```
>>> a = {1, 2, 3, 4}
>>> b = {3, 4, 5, 6}
>>> a & b
{3, 4}
>>> a - b
{1, 2}
>>> a | b
{1, 2, 3, 4, 5, 6}
>>> a ^ b
{1, 2, 5, 6}
```

Order warning

Set results are correct by membership, not by display order. Printed order may vary between runs.

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Code Example: python and telusko

Creating a set from a string stores each unique character once. Your example compares characters from "python" and "telusko".

Provided code sample

```
set1 = set("python")
set2 = set("telusko")

print(set1 & set2) # common characters
print(set1 - set2) # in set1, not in set2
print(set2 - set1) # in set2, not in set1
print(set1 ^ set2) # not common to both
print(set1 | set2) # in either set or both
```

Expression	Meaning	Members in result
set1 & set2	Common characters.	{o, t}
set1 - set2	Characters only in python.	{p, y, h, n}
set2 - set1	Characters only in telusko.	{e, l, u, s, k}
set1 ^ set2	Characters in one set only.	{p, y, h, n, e, l, u, s, k}
set1 set2	All unique characters.	{p, y, t, h, o, n, e, l, u, s, k}

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Membership Testing

Use the `in` keyword to quickly check whether a value exists inside a set.

Membership examples

```
>>> languages = {"python", "java", "c"}
>>> "python" in languages
True
>>> "ruby" in languages
False
```

Membership rule

Sets are designed for fast yes-or-no membership checks, especially when duplicates do not matter.

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Single-Element Tuple Tip

The lesson also clears up a tuple confusion: a one-value tuple needs a comma, otherwise Python treats the value as a normal integer.

Tuple reminder

```
>>> type((45))
<class 'int'>
>>> type((45,))
<class 'tuple'>
```

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Set Recap

Sets are useful when uniqueness and fast lookup matter more than position or order.

Feature	Set	List or tuple
Duplicates	Stores unique values only.	Can keep duplicates.
Order	Unordered display and storage.	List and tuple are ordered.
Empty value	Use set().	[] for list, () for tuple.
Best use	Membership and set math.	Sequence-style data.

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Quick Practice

Practice example

```
>>> a = {2, 4, 6, 8}
>>> b = {1, 2, 3, 4}
>>> a & b
{2, 4}
>>> a - b
{6, 8}
```

Day 11/55 takeaway

Sets store unique unordered values. Use them for fast membership tests, removing duplicates, and mathematical set operations.